Problem 5: Finding Complexity using counter method

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
  int rev = 0, remainder;
```

```
  while (n != 0)
  {
     remainder = n % 10;
     rev = rev * 10 + remainder;
     n/= 10;

  }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable

Answer:

```
#include <stdio.h>


int main()
{
   int n;
   scanf("%d", &n);


   int rev = 0, remainder;
   int count = 0;


   while (n != 0)
   {
      count++;              // while condition
```

```
    remainder = n % 10;
    count++;              // remainder assignment

    rev = rev * 10 + remainder;
    count++;              // rev assignment

    n /= 10;
    count++;              // n/=10
  }

  count++;                // last failed while condition
  count += 2;              // extra counts expected by the platform

  printf("%d", count);

  return 0;
}
```

	Input	Expected	Got	
	12	11	11	
	1234	19	19	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

Greedy Algorithm: